1) a)

$(1+\mathrm{i})^3 = 1 + 3\mathrm{i} + 3\mathrm{i}^2 + \mathrm{i}^3$
$= 1 + 3\mathrm{i} - 3 - \mathrm{i}$
$= -2 + 2\mathrm{i}$

1) b)

$(1+\mathrm{i})^4 = 1 + 4\mathrm{i} + 6\mathrm{i}^2 + 4\mathrm{i}^3 + \mathrm{i}^4$
$= 1 + 4\mathrm{i} - 6 - 4\mathrm{i} + 1$
$= -4$

1) c)

$(1+\mathrm{i})^5 = 1 + 5\mathrm{i} + 10\mathrm{i}^2 + 10\mathrm{i}^3 + 5\mathrm{i}^4 + \mathrm{i}^5$
$= 1 + 5\mathrm{i} - 10 - 10\mathrm{i} + 5 + \mathrm{i}$
$= -4 - 4\mathrm{i}$

1) d)

$(1+\mathrm{i})^6 = 1 + 6\mathrm{i} + 15\mathrm{i}^2 + 20\mathrm{i}^3 + 15\mathrm{i}^4 + 6\mathrm{i}^5 + \mathrm{i}^6$
$= 1 + 6\mathrm{i} - 15 - 20\mathrm{i} + 15 + 6\mathrm{i} - 1$
$= -8\mathrm{i}$

2) a)

$(a+b\mathrm{i})^3 = a^3 + 3a^2b\mathrm{i} + 3ab^2\mathrm{i}^2 + b^3\mathrm{i}^3$
$= a^3 + 3a^2b\mathrm{i} - 3ab^2 - b^3\mathrm{i}$
$= (a^3 - 3ab^2) + (3a^2b - b^3)\mathrm{i}$

2) b)

$(a+b\mathrm{i})^4 = a^4 + 4a^3b\mathrm{i} + 6a^2b^2\mathrm{i}^2 + 4ab^3\mathrm{i}^3 + b^4\mathrm{i}^4$
$= a^4 + 4a^3b\mathrm{i} - 6a^2b^2 - 4ab^3\mathrm{i} + b^4$
$= (a^4 - 6a^2b^2 + b^4) + (4a^3b - 4ab^3)\mathrm{i}$

2) c)

$(a+b\mathrm{i})^5 = a^5 + 5a^4b\mathrm{i} + 10a^3b^2\mathrm{i}^2 + 10a^2b^3\mathrm{i}^3 + 5ab^4\mathrm{i}^4 + b^5\mathrm{i}^5$
$= a^5 + 5a^4b\mathrm{i} - 10a^3b^2 - 10a^2b^3\mathrm{i} + 5ab^4 + b^5\mathrm{i}$
$= (a^5 - 10a^3b^2 + 5ab^4) + (5a^4b - 10a^2b^3 + b^5)\mathrm{i}$

3) a)

$(\cos\theta + \mathrm{i}\sin\theta)^2 = \cos^2\theta + 2\mathrm{i}\cos\theta\sin\theta - \sin^2\theta$

3) b)

Since $\cos 2\theta + \mathrm{i}\sin 2\theta = (\cos\theta + \mathrm{i}\sin\theta)^2$

Equate real parts:

$\cos 2\theta = \cos^2\theta - \sin^2\theta$ as req.

Equate imaginary parts:

$\sin 2\theta = 2\cos\theta\sin\theta$ as req.

4) a)

$(\cos\theta + \mathrm{i}\sin\theta)^3 = \cos^3\theta + 3\mathrm{i}\cos^2\theta\sin\theta - 3\cos\theta\sin^2\theta - \mathrm{i}\sin^3\theta$

4) b)

Since $\cos 3\theta + \mathrm{i}\sin 3\theta = (\cos\theta + \mathrm{i}\sin\theta)^3$

Equate real parts:

$\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta$

$= \cos^3\theta - 3\cos\theta(1 - \cos^2\theta)$

$= \cos^3\theta - 3\cos\theta + 3\cos^3\theta$

$= 4\cos^3\theta - 3\cos\theta$ as req.

Equate imaginary parts:

$\sin 3\theta = 3\cos^2\theta\sin\theta - \sin^3\theta$

$= 3(1 - \sin^2\theta)\sin\theta - \sin^3\theta$

$= 3\sin\theta - 3\sin^3\theta - \sin^3\theta$

$= 3\sin\theta - 4\sin^3\theta$ as req.

4) c)

$\tan 3\theta = \frac{\sin 3\theta}{\cos 3\theta} = \frac{3\sin\theta - 4\sin^3\theta}{4\cos^3\theta - 3\cos\theta}$

Divide numerator and denominator by $\cos^3\theta$:

$= \frac{3\tan\theta\sec^2\theta - 4\tan^3\theta}{4 - 3\sec^2\theta}$

Use the identity $1 + \tan^2\theta = \sec^2\theta$:

$= \frac{3\tan\theta(1+\tan^2\theta) - 4\tan^3\theta}{4 - 3(1+\tan^2\theta)}$

$= \frac{3\tan\theta + 3\tan^3\theta - 4\tan^3\theta}{4 - 3 - 3\tan^2\theta}$

$= \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$

5) a)

Let $f(x) = 4x^3 - 3x - 1$

$f(1) = 4 - 3 - 1 \therefore (x-1)$ is a factor

$4x^3 - 3x - 1 = (x-1)(4x^2 + 4x + 1) = (x-1)(2x+1)^2$

So $x = 1$ or $x = -\frac{1}{2}$

5) b)

$\cos 3\theta = 1 \Rightarrow 3\theta = 0$ or $3\theta = 2\pi$ so $\theta = 0$ or $\theta = \frac{2\pi}{3}$

5) c)

If $x = \cos\theta$ then $x = \cos 0$ or $x = \cos\frac{2\pi}{3}$

So $x = 1$ or $x = -\frac{1}{2}$

6) a)

$(\cos\theta + \mathrm{i}\sin\theta)^4 = \cos^4\theta + 4\mathrm{i}\cos^3\theta\sin\theta - 6\cos^2\theta\sin^2\theta - 4\mathrm{i}\cos\theta\sin^3\theta + \sin^4\theta$

Since $\cos 4\theta + \mathrm{i}\sin 4\theta = (\cos\theta + \mathrm{i}\sin\theta)^4$

Equate real parts:

$\cos 4\theta = \cos^4\theta - 6\cos^2\theta\sin^2\theta + \sin^4\theta$

$= \cos^4\theta - 6\cos^2\theta(1 - \cos^2\theta) + (1 - \cos^2\theta)(1 - \cos^2\theta)$

$= \cos^4\theta - 6\cos^2\theta + 6\cos^4\theta + 1 - 2\cos^2\theta + \cos^4\theta$

$= 8\cos^4\theta - 8\cos^2\theta + 1$ as req.

6) b)

$x^4 - x^2 = -\frac{1}{4} \Rightarrow 8x^4 - 8x^2 = -2 \Rightarrow 8x^4 - 8x^2 + 1 = -1$

Since $\cos 4\theta = 8\cos^4\theta - 8\cos^2\theta + 2$ then let $x = \cos\theta$

$\cos 4\theta = -1$ has solutions $\theta \in \left\{\ldots, \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}, \ldots\right\}$

Choose distinct values:

$x = \cos\frac{\pi}{4} = \frac{\sqrt{2}}{2}$ or $x = \cos\frac{3\pi}{4} = -\frac{\sqrt{2}}{2}$

$\therefore x = \frac{\sqrt{2}}{2}$ or $x = -\frac{\sqrt{2}}{2}$

7) a)

$(\cos\theta + \mathrm{i}\sin\theta)^4 = \cos^4\theta + 4\mathrm{i}\cos^3\theta\sin\theta - 6\cos^2\theta\sin^2\theta - 4\mathrm{i}\cos\theta\sin^3\theta + \sin^4\theta$

Since $\cos 4\theta + \mathrm{i}\sin 4\theta = (\cos\theta + \mathrm{i}\sin\theta)^4$

Equate imaginary parts:

$\sin 4\theta = 4\cos^3\theta\sin\theta - 4\cos\theta\sin^3\theta$ as req.

7) b)

$$\int_0^{\frac{3\pi}{4}} \cos^3\theta\sin\theta - \cos\theta\sin^3\theta \ \mathrm{d}\theta = \int_0^{\frac{3\pi}{4}} \frac{1}{4}\sin 4\theta \ \mathrm{d}\theta$$

$$= \left[-\frac{1}{16}\cos 4\theta\right]_0^{\frac{3\pi}{4}} = -\frac{1}{16}\cos\left(4\times\frac{3\pi}{4}\right) - \left(-\frac{1}{16}\cos(4\times 0)\right) = \frac{1}{16} - -\frac{1}{16} = \frac{1}{8}$$

as req.

8)

$(\cos\theta + \mathrm{i}\sin\theta)^3 = \cos^3\theta + 3\mathrm{i}\cos^2\theta\sin\theta - 3\cos\theta\sin^2\theta - \mathrm{i}\sin^3\theta$

Since $\cos 3\theta + \mathrm{i}\sin 3\theta = (\cos\theta + \mathrm{i}\sin\theta)^3$

Equate real parts:

$\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta$

$= \cos^3\theta - 3\cos\theta(1 - \cos^2\theta)$

$= \cos^3\theta - 3\cos\theta + 3\cos^3\theta$

$= 4\cos^3\theta - 3\cos\theta$

Equate imaginary parts:

$\sin 3\theta = 3\cos^2\theta\sin\theta - \sin^3\theta$

$= 3(1 - \sin^2\theta)\sin\theta - \sin^3\theta$

$= 3\sin\theta - 3\sin^3\theta - \sin^3\theta$

$= 3\sin\theta - 4\sin^3\theta$

So that

$\cot 3\theta = \frac{\cos 3\theta}{\sin 3\theta} = \frac{4\cos^3\theta - 3\cos\theta}{3\sin\theta - 4\sin^3\theta}$

Divide numerator and denominator by $\sin^3\theta$:

$= \frac{4\cot^3\theta - 3\cot\theta\,\mathrm{cosec}^2\theta}{3\,\mathrm{cosec}^2\theta - 4}$

Use the identity $1 + \cot^2\theta = \sec^2\theta$:

$= \frac{4\cot^3\theta - 3\cot\theta(1 + \cot^2\theta)}{3(1 + \cot^2\theta) - 4}$

$= \frac{4\cot^3\theta - 3\cot\theta - 3\cot^3\theta}{3 + 3\cot^2\theta - 4}$

$= \frac{\cot^3\theta - 3\cot\theta}{3\cot^2\theta - 1}$ as req.

9) a)

$(\cos\theta + \mathrm{i}\sin\theta)^5$

$= \cos^5\theta + 5\mathrm{i}\cos^4\theta\sin\theta - 10\cos^3\theta\sin^2\theta - 10\mathrm{i}\cos^2\theta\sin^3\theta + 5\cos\theta\sin^4\theta + \mathrm{i}\sin^5\theta$

Since $\cos 5\theta + \mathrm{i}\sin 5\theta = (\cos\theta + \mathrm{i}\sin\theta)^5$

Equate real parts:

$\cos 5\theta = \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta$

$= \cos^5\theta - 10\cos^3\theta(1 - \cos^2\theta) + 5\cos\theta(1 - \cos^2\theta)(1 - \cos^2\theta)$

$= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta(1 - 2\cos^2\theta + \cos^4\theta)$

$= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta - 10\cos^3\theta + 5\cos^5\theta$

$= 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$ as req.

9) b)

Equate imaginary parts:

$\sin 5\theta = 5\cos^4\theta \sin\theta - 10\cos^2\theta \sin^3\theta + \mathrm{i}\sin^5\theta$

$= 5(1-\sin^2\theta)(1-\sin^2\theta)\sin\theta - 10(1-\sin^2\theta)\sin^3\theta + \sin^5\theta$

$= 5\sin\theta\,(1-2\sin^2\theta+\sin^4\theta) - 10\sin^3\theta + 10\sin^5\theta + \sin^5\theta$

$= 5\sin\theta - 10\sin^3\theta + 5\sin^5\theta - 10\sin^3\theta + 10\sin^5\theta + \sin^5\theta$

$= 16\sin^5\theta - 20\sin^3\theta + 5\sin\theta$ as req.

9) c)

$(1+\tan^2\theta)(1+\tan^2\theta) = 1 + 2\tan^2\theta + \tan^4\theta$

9) d)

$\tan 5\theta = \frac{\sin 5\theta}{\cos 5\theta} = \frac{16\sin^5\theta - 20\sin^3\theta + 5\sin\theta}{16\cos^5\theta - 20\cos^3\theta + 5\cos\theta}$

Divide numerator and denominator by $\cos^5\theta$:

$= \frac{16\tan^5\theta - 20\tan^3\theta\sec^2\theta + 5\tan\theta\sec^4\theta}{16 - 20\sec^2\theta + 5\sec^4\theta}$

$= \frac{16\tan^5\theta - 20\tan^3\theta(1+\tan^2\theta) + 5\tan\theta(1+\tan^2\theta)(1+\tan^2\theta)}{16 - 20(1+\tan^2\theta) + 5(1+\tan^2\theta)(1+\tan^2\theta)}$

$= \frac{16\tan^5\theta - 20\tan^3\theta - 20\tan^5\theta + 5\tan\theta(1+2\tan^2\theta+\tan^4\theta)}{16 - 20 - 20\tan^2\theta + 5(1+2\tan^2\theta+\tan^4\theta)}$

$= \frac{16\tan^5\theta - 20\tan^3\theta - 20\tan^5\theta + 5\tan\theta + 10\tan^3\theta + 5\tan^5\theta}{16 - 20 - 20\tan^2\theta + 5 + 10\tan^2\theta + 5\tan^4\theta}$

$= \frac{\tan^5\theta - 10\tan^3\theta + 5\tan\theta}{1 - 10\tan^2\theta + 5\tan^4\theta}$ as req.

9) e)

$1 - 10x^2 + 5x^4 = x^5 - 10x^3 + 5x$

$\Rightarrow \frac{x^5 - 10x^3 + 5x}{1 - 10x^2 + 5x^4} = 1$

Let $x = \tan\theta$ so that $\tan(5\theta) = 1$

$\tan 5\theta = 1$ has solutions $\theta = \frac{\pi}{20} + \frac{k\pi}{5}$ for integer k

Choose distinct values:

$x = \tan\frac{\pi}{20}$ or $x = \tan\frac{\pi}{4}$ or $x = \tan\frac{9\pi}{20}$ or $x = \tan\frac{13\pi}{20}$ or $x = \tan\frac{17\pi}{20}$

$\therefore x = 0.158$ or $x = 1$ or $x = 6.314$ or $x = -1.963$ or $x = -0.510$

10) a)

$(1-\cos^2\theta)^3 = 1-3\cos^2\theta+3\cos^4\theta-\cos^6\theta$

10) b)

$(\cos\theta+\mathrm{i}\sin\theta)^6$

$=\cos^6\theta+6\mathrm{i}\cos^5\theta\sin\theta-15\cos^4\theta\sin^2\theta-20\mathrm{i}\cos^3\theta\sin^3\theta+15\cos^2\theta\sin^4\theta+6\mathrm{i}\cos\theta\sin^5\theta-\sin^6\theta$

Since $\cos 6\theta+\mathrm{i}\sin 6\theta=(\cos\theta+\mathrm{i}\sin\theta)^6$

Equate real parts:

$\cos 6\theta=\cos^6\theta-15\cos^4\theta\sin^2\theta+15\cos^2\theta\sin^4\theta-\sin^6\theta$

$=\cos^6\theta-15\cos^4\theta(1-\cos^2\theta)+15\cos^2\theta(1-\cos^2\theta)^2-(1-\cos^2\theta)^3$

$=\cos^6\theta-15\cos^4\theta+15\cos^6\theta+15\cos^2\theta(1-2\cos^2\theta+\cos^4\theta)-(1-3\cos^2\theta+3\cos^4\theta-\cos^6\theta)$

$=\cos^6\theta-15\cos^4\theta+15\cos^6\theta+15\cos^2\theta-30\cos^4\theta+15\cos^6\theta-1+3\cos^2\theta-3\cos^4\theta+\cos^6\theta$

$=32\cos^6\theta-48\cos^4\theta+18\cos^2\theta-1$ as req.

10) c)

$64x^6-96x^4+36x^2-3=0$

$\Rightarrow 64x^6-96x^4+36x^2-2=1$

$\Rightarrow 32x^6-48x^4+18x^2-1=\frac{1}{2}$

Let $x=\cos\theta$ so that $\cos(6\theta)=\frac{1}{2}$

$\cos 6\theta=\frac{1}{2}$ has solutions $\theta=\frac{\pi}{18}+\frac{k\pi}{3}$ for integer k and $\theta=-\frac{\pi}{18}+\frac{k\pi}{3}$ for integer k

Choose distinct values:

$x=\cos\frac{\pi}{18}$ or $x=\cos\frac{7\pi}{18}$ or $x=\cos\frac{13\pi}{18}$ or $x=\cos\frac{5\pi}{18}$ or $x=\cos\frac{11\pi}{18}$ or $x=\cos\frac{17\pi}{18}$

$\therefore x=0.9848$ or $x=0.3420$ or $x=-0.6428$ or $x=0.6428$ or $x=-0.3420$

or $x=-0.9848$ (4 DP)

11)

$(\cos\theta + \mathrm{i}\sin\theta)^6$

$= \cos^6\theta + 6\mathrm{i}\cos^5\theta\sin\theta - 15\cos^4\theta\sin^2\theta - 20\mathrm{i}\cos^3\theta\sin^3\theta + 15\cos^2\theta\sin^4\theta + 6\mathrm{i}\cos\theta\sin^5\theta - \sin^6\theta$

Since $\cos 6\theta + \mathrm{i}\sin 6\theta = (\cos\theta + \mathrm{i}\sin\theta)^6$

Equate imaginary parts:

$\sin 6\theta = 6\cos^5\theta\sin\theta - 20\cos^3\theta\sin^3\theta + 6\cos\theta\sin^5\theta$ as req.

12) a)

$(\cos\theta + \mathrm{i}\sin\theta)^4 = \cos^4\theta + 4\mathrm{i}\cos^3\theta\sin\theta - 6\cos^2\theta\sin^2\theta - 4\mathrm{i}\cos\theta\sin^3\theta + \sin^4\theta$

Since $\cos 4\theta + \mathrm{i}\sin 4\theta = (\cos\theta + \mathrm{i}\sin\theta)^4$

Equate imaginary parts: $\sin 4\theta = 4\cos^3\theta\sin\theta - 4\cos\theta\sin^3\theta$

So that

$\frac{\sin 4\theta}{\sin\theta} = \frac{4\cos^3\theta\sin\theta - 4\cos\theta\sin^3\theta}{\sin\theta}$

$= 4\cos^3\theta - 4\cos\theta\sin^2\theta$

Use the identity $\sin^2\theta + \cos^2\theta = 1$:

$= 4\cos^3\theta - 4\cos\theta(1 - \cos^2\theta)$

$= 4\cos^3\theta + 4\cos^3\theta - 4\cos\theta$

$= 8\cos^3\theta - 4\cos\theta$

12) b)

$\sin 4\theta = \sin\theta \Rightarrow \frac{\sin 4\theta}{\sin\theta} = 1 \Rightarrow 8\cos^3\theta - 4\cos\theta = 1$

$8\cos^3\theta - 4\cos\theta - 1 = 0$

Let $\mathrm{f}(x) = 8x^3 - 4x - 1$

$\mathrm{f}\left(-\frac{1}{2}\right) = -1 + 2 - 1 = 0 \therefore (2x+1)$ is a factor.

$8x^3 - 4x - 1 = (2x+1)(4x^2 - 2x - 1)$

So $8x^3 - 4x - 1 = 0$ has solutions of $x = -\frac{1}{2}$, $x = \frac{1+\sqrt{5}}{4}$ or $x = \frac{1-\sqrt{5}}{4}$

Using $x = \cos\theta$ then

$\cos\theta = -\frac{1}{2}$, $\cos\theta = \frac{1+\sqrt{5}}{4}$ or $\cos\theta = \frac{1-\sqrt{5}}{4}$

$\therefore \theta = \frac{2\pi}{3}$, $\theta = \frac{\pi}{5}$ or $\theta = \frac{3\pi}{5}$

$\therefore \theta = 2.09$, $\theta = 0.628$ or $\theta = 1.88$ (3 SF)

13) a)

$(\cos\theta + \mathrm{i}\sin\theta)^5$

$= \cos^5\theta + 5\mathrm{i}\cos^4\theta\sin\theta - 10\cos^3\theta\sin^2\theta - 10\mathrm{i}\cos^2\theta\sin^3\theta + 5\cos\theta\sin^4\theta + \mathrm{i}\sin^5\theta$

Since $\cos 5\theta + \mathrm{i}\sin 5\theta = (\cos\theta + \mathrm{i}\sin\theta)^5$

Equate real parts:

$\cos 5\theta = \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta$

$= \cos^5\theta - 10\cos^3\theta(1 - \cos^2\theta) + 5\cos\theta(1 - \cos^2\theta)(1 - \cos^2\theta)$

$= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta(1 - 2\cos^2\theta + \cos^4\theta)$

$= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta - 10\cos^3\theta + 5\cos^5\theta$

$= 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$

So that

$\frac{\cos 5\theta}{\cos\theta} = \frac{\cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta}{\cos\theta}$

$= \cos^4\theta - 10\cos^2\theta\sin^2\theta + 5\sin^4\theta$

Use the identity $\sin^2\theta + \cos^2\theta = 1$:

$= \cos^4\theta - 10\cos^2\theta(1 - \cos^2\theta) + 5(1 - \cos^2\theta)(1 - \cos^2\theta)$

$= \cos^4\theta - 10\cos^2\theta + 10\cos^4\theta + 5 - 10\cos^2\theta + 5\cos^4\theta$

$= 16\cos^4\theta - 20\cos^2\theta + 5$

13) b)

$\cos 5\theta = -\cos\theta \Rightarrow \frac{\cos 5\theta}{\cos\theta} = -1 \Rightarrow 16\cos^4\theta - 20\cos^2\theta + 5 = -1$

$16\cos^4\theta - 20\cos^2\theta + 6 = 0$

Let $\mathrm{f}(x) = 16x^4 - 20x^2 + 6$

$16x^4 - 20x^2 + 6 = 2(2x^2 - 1)(4x^2 - 3)$

So $16x^4 - 20x^2 + 6 = 0$ has solutions of $x = -\frac{1}{\sqrt{2}}$, $x = \frac{1}{\sqrt{2}}$, $x = -\frac{\sqrt{3}}{2}$ or $x = \frac{\sqrt{3}}{2}$

Using $x = \cos\theta$ then

$\cos\theta = -\frac{1}{\sqrt{2}}$, $\cos\theta = \frac{1}{\sqrt{2}}$, $\cos\theta = -\frac{\sqrt{3}}{2}$ or $\cos\theta = \frac{\sqrt{3}}{2}$

$\therefore \theta = \frac{3\pi}{4}$, $\theta = \frac{\pi}{4}$, $\theta = \frac{5\pi}{6}$ or $\theta = \frac{\pi}{6}$

14) a)

$z^3 = (1 + 3\mathrm{i})^3 = 1 + 9\mathrm{i} - 27 - 27\mathrm{i} = -26 - 18\mathrm{i}$

14) b)

$\arg z = \arctan\left(\frac{3}{1}\right) = \arctan 3$

14) c)

$(\cos\theta + \mathrm{i}\sin\theta)^3 = \cos^3\theta + 3\mathrm{i}\cos^2\theta\sin\theta - 3\cos\theta\sin^2\theta - \mathrm{i}\sin^3\theta$

Since $\cos 3\theta + \mathrm{i}\sin 3\theta = (\cos\theta + \mathrm{i}\sin\theta)^3$

Equate real parts:

$\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta$

$= \cos^3\theta - 3\cos\theta(1 - \cos^2\theta)$

$= \cos^3\theta - 3\cos\theta + 3\cos^3\theta$

$= 4\cos^3\theta - 3\cos\theta$ as req.

Equate imaginary parts:

$\sin 3\theta = 3\cos^2\theta\sin\theta - \sin^3\theta$

$= 3(1 - \sin^2\theta)\sin\theta - \sin^3\theta$

$= 3\sin\theta - 3\sin^3\theta - \sin^3\theta$

$= 3\sin\theta - 4\sin^3\theta$ as req.

14) d)

$\tan 3\theta = \frac{\sin 3\theta}{\cos 3\theta} = \frac{3\sin\theta - 4\sin^3\theta}{4\cos^3\theta - 3\cos\theta}$

Divide numerator and denominator by $\cos^3\theta$:

$= \frac{3\tan\theta\sec^2\theta - 4\tan^3\theta}{4 - 3\sec^2\theta}$

Use the identity $1 + \tan^2\theta = \sec^2\theta$:

$= \frac{3\tan\theta(1+\tan^2\theta) - 4\tan^3\theta}{4 - 3(1+\tan^2\theta)}$

$= \frac{3\tan\theta + 3\tan^3\theta - 4\tan^3\theta}{4 - 3 - 3\tan^2\theta}$

$= \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$ as req.

14) e)

From $z = 1 + 3\text{i}$ we have $\arg z = \theta = \arctan 3$

So that $\tan\theta = 3$

Using the identity, this gives $\tan 3\theta = \frac{3\times3-3^3}{1-3\times3^2} = -\frac{18}{-26} = \frac{9}{13}$

Also $z^3 = (1+3\text{i})^3 = -26 - 18\text{i}$

But since we know that for complex number $a + b\text{i}$, its argument ϕ satisfies $\tan\phi = \frac{b}{a}$

Using the argument, this gives $\tan 3\theta = \frac{\text{Im}(z^3)}{\text{Re}(z^3)} = \frac{-18}{-26} = \frac{9}{13}$

So the results are consistent.

15) a)

$(\cosh x + \sinh x)^3 = \cosh^3 x + 3\cosh^2 x \sinh x + 3\cosh x \sinh^2 x + \sinh^3 x$

15) b)

$(\cosh x + \sinh x)^3 = \left(\frac{e^x+e^{-x}}{2} + \frac{e^x-e^{-x}}{2}\right)^3$

$= \left(\frac{e^x}{2} + \frac{e^{-x}}{2} + \frac{e^x}{2} - \frac{e^{-x}}{2}\right)^3$

$= (e^x)^3 = e^{3x}$

$= \frac{e^{3x}}{2} + \frac{e^{-3x}}{2} + \frac{e^{3x}}{2} - \frac{e^{-3x}}{2}$

$= \frac{e^{3x}+e^{-3x}}{2} + \frac{e^{3x}-e^{-3x}}{2}$

$= \cosh 3x + \sinh 3x$

$(\cosh x - \sinh x)^3 = \left(\frac{e^x+e^{-x}}{2} - \frac{e^x-e^{-x}}{2}\right)^3$

$= \left(\frac{e^x}{2} + \frac{e^{-x}}{2} - \frac{e^x}{2} + \frac{e^{-x}}{2}\right)^3$

$= (e^{-x})^3 = e^{-3x}$

$= \frac{e^{3x}}{2} + \frac{e^{-3x}}{2} - \frac{e^{3x}}{2} + \frac{e^{-3x}}{2}$

$= \frac{e^{3x}+e^{-3x}}{2} - \frac{e^{3x}-e^{-3x}}{2}$

$= \cosh 3x - \sinh 3x$

So that $(\cosh x + \sinh x)^3 = \cosh 3x + \sinh 3x$ and

$(\cosh x - \sinh x)^3 = \cosh 3x - \sinh 3x$

Alternatively:

Since $\cosh x + \sinh x = \frac{1}{2}(e^x + e^{-x}) + \frac{1}{2}(e^x - e^{-x}) = e^x$

$(\cosh x + \sinh x)^3 = e^{3x} = \cosh(3x) + \sinh(3x)$

Since $\cosh x - \sinh x = \frac{1}{2}(e^x + e^{-x}) - \frac{1}{2}(e^x - e^{-x}) = e^{-x}$

$(\cosh x - \sinh x)^3 = e^{-3x} = \cosh(-3x) + \sinh(-3x)$

$= \cosh 3x - \sinh 3x$

So that $(\cosh x + \sinh x)^3 = \cosh 3x + \sinh 3x$ and

$(\cosh x - \sinh x)^3 = \cosh 3x - \sinh 3x$

15) c)

$(\cosh x + \sinh x)^3 + (\cosh x - \sinh x)^3 = \cosh 3x + \sinh 3x + \cosh 3x - \sinh(3x)$
$= 2\cosh 3x$
$\therefore 2\cosh 3x = \cosh^3 x + 3\cosh^2 x \sinh x + 3\cosh x \sinh^2 x + \sinh^3 x + (\cosh^3 x - 3\cosh^2 x \sinh x + 3\cosh x \sinh^2 x - \sinh^3 x)$
$= 2\cosh^3 x + 6\cosh x \sinh^2 x$
$\therefore \cosh 3x = \cosh^3 x + 3\cosh x \sinh^2 x$
$(\cosh x + \sinh x)^3 - (\cosh x - \sinh x)^3 = \cosh 3x + \sinh 3x - \cosh 3x + \sinh 3x$
$= 2\sinh 3x$
$\therefore 2\sinh 3x = \cosh^3 x + 3\cosh^2 x \sinh x + 3\cosh x \sinh^2 x + \sinh^3 x - (\cosh^3 x - 3\cosh^2 x \sinh x + 3\cosh x \sinh^2 x - \sinh^3 x)$
$= 6\cosh^2 x \sinh x + 2\sinh^3 x$
$\therefore \sinh 3x = 3\cosh^2 x \sinh x + \sinh^3 x$
Hence $\cosh 3x = \cosh^3 x + 3\cosh x \sinh^2 x$ and $\sinh 3x = 3\cosh^2 x \sinh x + \sinh^3 x$

15) d)

$\cosh 3x = \cosh^3 x + 3\cosh x \sinh^2 x = 2^3 + 3 \times 2 \times 3 = 26$ and
$\sinh 3x = 3\cosh^2 x \sinh x + \sinh^3 x = 3 \times 2^2 \times \sqrt{3} + 3\sqrt{3} = 15\sqrt{3}$
$\cosh 3x = 26$ and $\sinh 3x = 15\sqrt{3}$

16) a)

$e^{iz} = \cos z + i\sin z$
$\cosh(iz) = \frac{1}{2}\left(e^{iz} + e^{-iz}\right) = \frac{1}{2}(\cos z + i\sin z + \cos(-z) + i\sin(-z))$
$= \frac{1}{2}(\cos z + i\sin z + \cos z - i\sin z) = \frac{1}{2}(2\cos z) = \cos z$ as req.

16) b)

$\cosh\left(\frac{3\pi}{4}i\right) = \cos\frac{3\pi}{4} = -\frac{\sqrt{2}}{2}$

17) a)

$e^{iz} = \cos z + i\sin z$
$\sinh(iz) = \frac{1}{2}\left(e^{iz} - e^{-iz}\right) = \frac{1}{2}(\cos z + i\sin z - \cos(-z) - i\sin(-z))$
$= \frac{1}{2}(\cos z + i\sin z - \cos z + i\sin z) = \frac{1}{2}(2i\sin z) = i\sin z$ as req.

17) b)

$\sinh\left(\frac{5\pi}{6}\mathrm{i}\right) = \mathrm{i}\sin\frac{5\pi}{6} = \frac{1}{2}\mathrm{i}$

18) a)

$\mathrm{e}^{\mathrm{i}z} = \cos z + \mathrm{i}\sin z$

$\sinh(\mathrm{i}z) = \frac{1}{2}\left(\mathrm{e}^{\mathrm{i}z} - \mathrm{e}^{-\mathrm{i}z}\right) = \frac{1}{2}(\cos z + \mathrm{i}\sin z - \cos(-z) - \mathrm{i}\sin(-z))$

$= \frac{1}{2}(\cos z + \mathrm{i}\sin z - \cos z + \mathrm{i}\sin z) = \frac{1}{2}(2\mathrm{i}\sin z) = \mathrm{i}\sin z$ as req.

18) b)

$\frac{3\pi}{7} = 3\theta$ where $\theta = \frac{\pi}{7}$

Since $(\cos\theta + \mathrm{i}\sin\theta)^3 = \cos(3\theta) + \mathrm{i}\sin(3\theta)$

$\left(\cos\frac{\pi}{7} + \mathrm{i}\sin\frac{\pi}{7}\right)^3 = \cos\frac{3\pi}{7} + \mathrm{i}\sin\frac{3\pi}{7}$

Consider the LHS:

$\left(\cos\frac{\pi}{7} + \mathrm{i}\sin\frac{\pi}{7}\right)^3 = \cos^3\left(\frac{\pi}{7}\right) + 3\mathrm{i}\cos^2\left(\frac{\pi}{7}\right)\sin\frac{\pi}{7} + 3\cos\frac{\pi}{7}\sin^2\left(\frac{\pi}{7}\right) - \mathrm{i}\sin^3\left(\frac{\pi}{7}\right)$

Compare the imaginary parts:

$\mathrm{i}\sin\frac{3\pi}{7} = 3\mathrm{i}\cos^2\left(\frac{\pi}{7}\right)\sin\frac{\pi}{7} - \mathrm{i}\sin^3\left(\frac{\pi}{7}\right)$

But since $\sinh(\mathrm{i}z) = \mathrm{i}\sin z$ then $\sinh\left(\frac{3\pi}{7}\mathrm{i}\right) = 3\mathrm{i}\cos^2\left(\frac{\pi}{7}\right)\sin\frac{\pi}{7} - \mathrm{i}\sin^3\left(\frac{\pi}{7}\right)$

19)

Since $\cos 3\theta + \mathrm{i}\sin 3\theta = (\cos\theta + \mathrm{i}\sin\theta)^3$

Equate real parts:

$\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta$

$= \cos^3\theta - 3\cos\theta(1 - \cos^2\theta)$

$= \cos^3\theta - 3\cos\theta + 3\cos^3\theta$

$= 4\cos^3\theta - 3\cos\theta$

Considering $x = \cos\frac{4\pi}{9}$ then let $\theta = \frac{4\pi}{9}$ so that $3\theta = \frac{4\pi}{3}$

So $\cos 3\theta = \cos\frac{4\pi}{3} = -\frac{1}{2}$

So using $x = \cos\theta$ in $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$:

$\cos\frac{4\pi}{3} = 4x^3 - 3x$

$-\frac{1}{2} = 4x^3 - 3x$

$\therefore 8x^3 - 6x + 1 = 0$ as req.

20) a)

$(\cos\theta + \mathrm{i}\sin\theta)^5$

$= \cos^5\theta + 5\mathrm{i}\cos^4\theta\sin\theta - 10\cos^3\theta\sin^2\theta - 10\mathrm{i}\cos^2\theta\sin^3\theta + 5\cos\theta\sin^4\theta + \mathrm{i}\sin^5\theta$

Since $\cos 5\theta + \mathrm{i}\sin 5\theta = (\cos\theta + \mathrm{i}\sin\theta)^5$

Equate real parts:

$\cos 5\theta = \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta$

$= \cos^5\theta - 10\cos^3\theta(1 - \cos^2\theta) + 5\cos\theta(1 - \cos^2\theta)(1 - \cos^2\theta)$

$= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta(1 - 2\cos^2\theta + \cos^4\theta)$

$= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta - 10\cos^3\theta + 5\cos^5\theta$

$= 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$ as req.

20) b)

$16\cos^5\theta - 20\cos^3\theta + 5\cos\theta = 0$

Let $x = \cos\theta$ so that $16x^5 - 20x^3 + 5x = 0$

$16x^5 - 20x^3 + 5x = x(16x^4 - 20x^2 + 5)$

So that

$x = 0$ or $x^2 = \frac{20 \pm \sqrt{400-320}}{32} = \frac{20 \pm \sqrt{80}}{32} = \frac{20 \pm 4\sqrt{5}}{32} = \frac{5 \pm \sqrt{5}}{8}$

So that $\cos\theta = 0$ or $\cos^2\theta = \frac{5+\sqrt{5}}{8}$ or $\cos^2\theta = \frac{5-\sqrt{5}}{8}$

Solving $\cos 5\theta = 0$ for $0 \leq \theta \leq \pi$ gives $\frac{\pi}{10}$, $\frac{3\pi}{10}$, $\frac{5\pi}{10}$, $\frac{7\pi}{10}$ or $\frac{9\pi}{10}$

Since $\frac{\pi}{10}$ and $\frac{3\pi}{10}$ are both in the first quadrant, then for cosine, $\frac{\pi}{10}$ must give the larger solution and $\frac{3\pi}{10}$ must give the smaller solution.

So that

$\cos^2\left(\frac{\pi}{10}\right) = \frac{5+\sqrt{5}}{8}$ and $\cos^2\left(\frac{3\pi}{10}\right) = \frac{5-\sqrt{5}}{8}$ as req.

21) a)

$(\cos\theta + \mathrm{i}\sin\theta)^6$

$= \cos^6\theta + 6\mathrm{i}\cos^5\theta\sin\theta - 15\cos^4\theta\sin^2\theta - 20\mathrm{i}\cos^3\theta\sin^3\theta + 15\cos^2\theta\sin^4\theta + 6\mathrm{i}\cos\theta\sin^5\theta - \sin^6\theta$

Since $\cos 6\theta + \mathrm{i}\sin 6\theta = (\cos\theta + \mathrm{i}\sin\theta)^6$

Equate real parts:

$\cos 6\theta = \cos^6\theta - 15\cos^4\theta\sin^2\theta + 15\cos^2\theta\sin^4\theta - \sin^6\theta$

$= \cos^6\theta - 15\cos^4\theta(1-\cos^2\theta) + 15\cos^2\theta(1-\cos^2\theta)^2 - (1-\cos^2\theta)^3$

$= \cos^6\theta - 15\cos^4\theta + 15\cos^6\theta + 15\cos^2\theta(1 - 2\cos^2\theta + \cos^4\theta) - (1 - 3\cos^2\theta + 3\cos^4\theta - \cos^6\theta)$

$= \cos^6\theta - 15\cos^4\theta + 15\cos^6\theta + 15\cos^2\theta - 30\cos^4\theta + 15\cos^6\theta - 1 + 3\cos^2\theta - 3\cos^4\theta + \cos^6\theta$

$= 32\cos^6\theta - 48\cos^4\theta + 18\cos^2\theta - 1$ as req.

21) b)

$32\cos^6\theta - 48\cos^4\theta + 18\cos^2\theta - 1 = 0$

Let $x = \cos^2\theta$ so that $32x^3 - 48x^2 + 18x - 1 = 0$

Let $f(x) = 32x^3 - 48x^2 + 18x - 1$

$f\left(\frac{1}{2}\right) = 4 - 12 + 9 - 1 = 0 \therefore (2x - 1)$ is a factor

$32x^3 - 48x^2 + 18x - 1 = (2x - 1)(16x^2 - 16x + 1)$

So that

$x = \frac{1}{2}$ or $x = \frac{16 \pm \sqrt{256-64}}{32} = \frac{16 \pm \sqrt{192}}{32} = \frac{16 \pm 8\sqrt{3}}{32} = \frac{2 \pm \sqrt{3}}{4}$

So that $\cos^2\theta = \frac{1}{2}$ or $\cos^2\theta = \frac{2+\sqrt{3}}{4}$ or $\cos^2\theta = \frac{2-\sqrt{3}}{4}$

Solving $\cos(6\theta) = 0$ for $0 \leq \theta \leq \pi$ gives $\frac{\pi}{12}, \frac{3\pi}{12}, \frac{5\pi}{12}, \frac{7\pi}{12}$ or $\frac{9\pi}{12}$

Since $\frac{\pi}{12}$ and $\frac{5\pi}{12}$ are both in the first quadrant, then for cosine, $\frac{\pi}{12}$ must give the larger solution and $\frac{5\pi}{12}$ must give the smaller solution.

So that

$\cos^2\left(\frac{\pi}{12}\right) = \frac{2+\sqrt{3}}{4}$ and $\cos^2\left(\frac{5\pi}{12}\right) = \frac{2-\sqrt{3}}{4}$ as req.

22) a)

$(\cos\theta + \mathrm{i}\sin\theta)^5$

$= \cos^5\theta + 5\mathrm{i}\cos^4\theta\sin\theta - 10\cos^3\theta\sin^2\theta - 10\mathrm{i}\cos^2\theta\sin^3\theta + 5\cos\theta\sin^4\theta + \mathrm{i}\sin^5\theta$

Since $\cos 5\theta + \mathrm{i}\sin 5\theta = (\cos\theta + \mathrm{i}\sin\theta)^5$

Equate real parts:

$\cos 5\theta = \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta$

$= \cos^5\theta - 10\cos^3\theta(1 - \cos^2\theta) + 5\cos\theta(1 - \cos^2\theta)(1 - \cos^2\theta)$

$= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta(1 - 2\cos^2\theta + \cos^4\theta)$

$= \cos^5\theta - 10\cos^3\theta + 10\cos^5\theta + 5\cos\theta - 10\cos^3\theta + 5\cos^5\theta$

$= 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$

$\therefore \cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$

22) b)

Solving $\cos 5\theta = 1$:

$16\cos^5\theta - 20\cos^3\theta + 5\cos\theta = 1$

Let $x = \cos\theta$ so that $16x^5 - 20x^3 + 5x - 1 = 0$

$16x^5 - 20x^3 + 5x - 1 = (x-1)(16x^4 + 16x^3 - 4x^2 - 4x + 1)$

$= (x-1)(4x^2 + 2x - 1)^2$

$\therefore (x-1)(4x^2 + 2x - 1)^2 = 0$ as req.

22) c)

So that

$x = 1$ or $x = \frac{-2\pm\sqrt{4+16}}{8} = \frac{-2\pm 2\sqrt{5}}{8} = \frac{-1\pm\sqrt{5}}{4}$

So that $\cos\theta = 1$ or $\cos\theta = \frac{-1+\sqrt{5}}{4}$ or $\cos\theta = \frac{-1-\sqrt{5}}{4}$

Solving $\cos 5\theta = 1$ for $0 \leq \theta \leq 180°$ gives 0, 72°, or 144°

Since 72° is in the first quadrant and 144° is in the second quadrant, 72° must give the positive solution and 144° must give the negative solution.

$\therefore \cos 72° = \frac{-1+\sqrt{5}}{4}$ and $\cos 144° = \frac{-1-\sqrt{5}}{4}$ as req.

23) a)

Since $\cos 2A = 1 - 2\sin^2 A$ then $\cos\theta = 1 - 2\sin^2\left(\frac{\theta}{2}\right)$

Replace θ with $n\theta$:

$\cos n\theta = 1 - 2\sin^2\left(\frac{n\theta}{2}\right)$

23) b)

Since $\sin 2A = 2\sin A\cos A$ then $\sin\theta = 2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)$

Replace θ with $n\theta$:

$\sin n\theta = 2\sin\left(\frac{n\theta}{2}\right)\cos\left(\frac{n\theta}{2}\right)$

23) c)

$$\sum_{r=1}^{n}\cos(r\theta) = \mathrm{Re}[(\cos\theta + \mathrm{i}\sin\theta) + (\cos\theta + \mathrm{i}\sin\theta)^2 + \cdots + (\cos\theta + \mathrm{i}\sin\theta)^n]$$

This is a geometric series with $a = \cos\theta + \mathrm{i}\sin\theta$ and $r = \cos\theta + \mathrm{i}\sin\theta$

So use $S_n = \frac{a(1-r^n)}{1-r}$

$$\frac{(\cos\theta + \mathrm{i}\sin\theta)(1-(\cos\theta + \mathrm{i}\sin\theta)^n)}{1-(\cos\theta + \mathrm{i}\sin\theta)} = \frac{(\cos\theta + \mathrm{i}\sin\theta)(1-\cos n\theta - \mathrm{i}\sin n\theta)}{1-\cos\theta - \mathrm{i}\sin\theta}$$

Rationalise the denominator:

$$\frac{(\cos\theta + \mathrm{i}\sin\theta)(1-\cos n\theta - \mathrm{i}\sin n\theta)}{1-\cos\theta - \mathrm{i}\sin\theta} \times \frac{1-\cos\theta + \mathrm{i}\sin\theta}{1-\cos\theta + \mathrm{i}\sin\theta}$$

Which gives a denominator of

$$(1-\cos\theta - \mathrm{i}\sin\theta)(1-\cos\theta + \mathrm{i}\sin\theta) = (1-\cos\theta)^2 + \sin^2\theta$$

$$= 1 - 2\cos\theta + \cos^2\theta + \sin^2\theta = 2 - 2\cos\theta$$

So this gives:

$$\frac{(\cos\theta + \mathrm{i}\sin\theta)(1-\cos n\theta - \mathrm{i}\sin n\theta)(1-\cos\theta + \mathrm{i}\sin\theta)}{2-2\cos\theta}$$

Expand brackets:

$$(\cos\theta + \mathrm{i}\sin\theta)(1-\cos\theta + \mathrm{i}\sin\theta)$$

$$= \cos\theta - \cos^2\theta + \mathrm{i}\cos\theta\sin\theta + \mathrm{i}\sin\theta - \mathrm{i}\cos\theta\sin\theta - \sin^2\theta$$

$$= \cos\theta + \mathrm{i}\sin\theta - 1$$

So that the fraction becomes:

$$\frac{(\cos\theta + \mathrm{i}\sin\theta - 1)(1-\cos n\theta - \mathrm{i}\sin n\theta)}{2-2\cos\theta}$$

Expand brackets:

$$(\cos\theta + \mathrm{i}\sin\theta - 1)(1-\cos n\theta - \mathrm{i}\sin n\theta)$$

$$= \cos\theta - \cos\theta\cos n\theta - \mathrm{i}\cos\theta\sin n\theta + \mathrm{i}\sin\theta - \mathrm{i}\sin\theta\cos n\theta + \sin\theta\sin n\theta - 1 + \cos n\theta + \mathrm{i}\sin n\theta$$

Considering only the real parts:

$$\sum_{r=1}^{n}\cos(r\theta) = \frac{\cos\theta - \cos\theta\cos n\theta + \sin\theta\sin n\theta - 1 + \cos n\theta}{2-2\cos\theta}$$

$$= \frac{\cos\theta - 1 - \cos\theta\cos n\theta + \sin\theta\sin n\theta + \cos n\theta}{2-2\cos\theta}$$

$$= \frac{(\cos\theta - 1)(1-\cos n\theta) + \sin\theta\sin n\theta}{2-2\cos\theta}$$

We know $\cos\theta = 1 - 2\sin^2\left(\frac{\theta}{2}\right)$ and $\sin\theta = 2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)$,

and also $\cos n\theta = 1 - 2\sin^2\left(\frac{n\theta}{2}\right)$ and $\sin n\theta = 2\sin\left(\frac{n\theta}{2}\right)\cos\left(\frac{n\theta}{2}\right)$

So that:

$\cos\theta - 1 = -2\sin^2\left(\frac{\theta}{2}\right)$

$1 - \cos n\theta = 2\sin^2\left(\frac{n\theta}{2}\right)$

$2 - 2\cos\theta = 4\sin^2\left(\frac{\theta}{2}\right)$

$\sin\theta\sin n\theta = 4\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)\sin\left(\frac{n\theta}{2}\right)\cos\left(\frac{n\theta}{2}\right)$

Consider the numerator only:

$(\cos\theta - 1)(1 - \cos n\theta) + \sin\theta\sin n\theta$

$= -4\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{n\theta}{2}\right) + 4\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)\sin\left(\frac{n\theta}{2}\right)\cos\left(\frac{n\theta}{2}\right)$

$= 4\sin\left(\frac{\theta}{2}\right)\sin\left(\frac{n\theta}{2}\right)\left[-\sin\left(\frac{\theta}{2}\right)\sin\left(\frac{n\theta}{2}\right) + \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{n\theta}{2}\right)\right]$

Using the compound angle formula, $\cos(A+B) = \cos A\cos B - \sin A\sin B$

$= 4\sin\left(\frac{\theta}{2}\right)\sin\left(\frac{n\theta}{2}\right)\cos\left(\frac{\theta}{2} + \frac{n\theta}{2}\right)$

Therefore

$$\sum_{r=1}^{n}\cos(r\theta) = \frac{4\sin\left(\frac{\theta}{2}\right)\sin\left(\frac{n\theta}{2}\right)\cos\left(\frac{\theta}{2} + \frac{n\theta}{2}\right)}{4\sin^2\left(\frac{\theta}{2}\right)}$$

$$= \frac{\sin\left(\frac{n\theta}{2}\right)\cos\left(\frac{(n+1)\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}$$

As req.

23) d)

Similarly:

Considering only the imaginary parts:

$$\sum_{r=1}^{n} \sin(r\theta) = \frac{-\cos\theta \sin n\theta + \sin\theta - \sin\theta \cos n\theta + \sin n\theta}{2 - 2\cos\theta}$$

$$= \frac{(1 - \cos\theta)\sin n\theta + \sin\theta\,(1 - \cos n\theta)}{2 - 2\cos\theta}$$

Since

$$1 - \cos\theta = 2\sin^2\left(\frac{\theta}{2}\right)$$

$$1 - \cos n\theta = 2\sin^2\left(\frac{n\theta}{2}\right)$$

$$2 - 2\cos\theta = 4\sin^2\left(\frac{\theta}{2}\right)$$

This gives

$$\frac{2\sin^2\left(\frac{\theta}{2}\right)\sin n\theta + 2\sin\theta\sin^2\left(\frac{n\theta}{2}\right)}{4\sin^2\left(\frac{\theta}{2}\right)}$$

$$= \frac{\sin^2\left(\frac{\theta}{2}\right)\sin n\theta + \sin\theta\sin^2\left(\frac{n\theta}{2}\right)}{2\sin^2\left(\frac{\theta}{2}\right)}$$

Since $\sin\theta = 2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)$ and $\sin n\theta = 2\sin\left(\frac{n\theta}{2}\right)\cos\left(\frac{n\theta}{2}\right)$

$$= \frac{2\sin^2\left(\frac{\theta}{2}\right)\sin\left(\frac{n\theta}{2}\right)\cos\left(\frac{n\theta}{2}\right) + 2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)\sin^2\left(\frac{n\theta}{2}\right)}{2\sin^2\left(\frac{\theta}{2}\right)}$$

$$= \frac{\sin\left(\frac{\theta}{2}\right)\sin\left(\frac{n\theta}{2}\right)\cos\left(\frac{n\theta}{2}\right) + \cos\left(\frac{\theta}{2}\right)\sin^2\left(\frac{n\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}$$

$$= \frac{\sin\left(\frac{n\theta}{2}\right)\left[\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{n\theta}{2}\right) + \cos\left(\frac{\theta}{2}\right)\sin\left(\frac{n\theta}{2}\right)\right]}{\sin\left(\frac{\theta}{2}\right)}$$

Using the compound angle formula, $\sin(A + B) = \sin A\cos B + \cos A\sin B$

$$\sum_{r=1}^{n} \sin(r\theta) = \frac{\sin\left(\frac{n\theta}{2}\right)\sin\left(\frac{\theta}{2} + \frac{n\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}$$

$$= \frac{\sin\left(\frac{n\theta}{2}\right)\sin\left(\frac{(n+1)\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}$$

As req.